Type MS · Describe the steps of IVF treatment

Answer 1 of 4 · Question 6 part c · 4 marks

2024 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

Question number			
6 (a)	<table><tr><th rowspan="2">Hormone</th><th colspan="4">Function of hormone</th></tr><tr><th>causes ovulation</th><th>builds up uterus lining after menstruation</th><th>maintains uterus lining during pregnancy</th><th>stimulates a follicle to mature</th></tr><tr><td>progesterone</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>FSH</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>oestrogen</td><td></td><td>✓</td><td></td><td></td></tr></table>	One mark for each row Reject more than one tick in each row	(3)
(b)	condom/ diaphragm/femidom/cap;		(1)
(c)	Any four from the following • female given FSH and LH; • egg and sperms collected/extracted/removed; • egg fertilised by sperm in laboratory/ glass dish/petri dish.; • embryo formed/ develops; • embryo inserted into uterus;		(4)
(d)	Any two from the following • passive immunity/antibodies passed from mother to baby/less chance of infection (for baby); • increases (emotional)bonding; • breast milk contains a balance/composition/combination of nutrients; • contains hormones to help development (of baby);		(2)

Total for Question 6 = 10 marks

Type MS · Describe the steps of IVF treatment

Answer 2 of 4 · Question 5 part b.i · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

5	(a)	(i)	ovulation;		1
		(ii)	fertilisation		1
		(iii)	Any two from: • zygote/embryo/cells divides/splits; • each half develops into one offspring;		1 1
	(b)	(i)	**A** FSH and LH;		1
		(ii)	Any four from: • removal of egg from female ovary; • collect sperm from male; • egg fertilised; • embryos formed; • embryo inserted into uterus/female;	Allow egg and sperm fuse Allow zygote formed	Max 4

Total question 5 = 9 marks

Type MS · Describe the steps of IVF treatment

Answer 2 of 4 · Question 5 part b.i · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe the steps of IVF treatment

Answer 3 of 4 · Question 5 part b.ii · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

5	(a)	(i)	ovulation;		1
		(ii)	fertilisation		1
		(iii)	Any two from: • zygote/embryo/cells divides/splits; • each half develops into one offspring;		1 1
	(b)	(i)	**A** FSH and LH;		1
		(ii)	Any four from: • removal of egg from female ovary; • collect sperm from male; • egg fertilised; • embryos formed; • embryo inserted into uterus/female;	Allow egg and sperm fuse Allow zygote formed	Max 4

Total question 5 = 9 marks

Type MS · Describe the steps of IVF treatment

Answer 3 of 4 · Question 5 part b.ii · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe the steps of IVF treatment

Answer 4 of 4 · Question 8 part a.ii · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
8	(a)	(i)	triggers ovulation/causes release of the egg;		1
		(ii)	Three from: • FSH stimulates growth/production of follicles in the ovary; • more eggs mature/produced; • increases the chances of more than one egg being released; • each egg is fertilised by a sperm;		3
	(b)	(i)	Three from: • placenta supplies glucose/oxygen from the mother; • twins have to share the supply of glucose/receive less glucose/oxygen; • less aerobic respiration; • less energy is released in the foetus and less growth occurs;	Allow nutrients for mp's 1 and 2	3
		(ii)	Two from • same egg and sperm used; • zygote splits/divides • forming 2 embryos;		2
	(c)	(i)	75th		1
		(ii)	• baby is weighed regularly; • growth should follow a percentile/graph/compare with (growth) chart/graph/compare mass to others of the same age; • detect if the baby gains or loses too much mass;		3

Total 13 marks

Type MS · Describe the steps of IVF treatment

Answer 4 of 4 · Question 8 part a.ii · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom